Lecture Notes

BAYESIAN ANALYSIS

DSA 8505

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Contents

5 Bayesian Hypothesis Testing and Model Averaging

5.1 Introduction to Bayesian Hypothesis Testing

Bayesian hypothesis testing is a framework for statistical inference that uses probabilities to quantify uncertainty about hypotheses. Unlike frequentist methods, which rely on p-values and rejection regions, Bayesian methods incorporate prior information and update beliefs through Bayes' theorem.

5.1.1 The Bayesian Paradigm

The Bayesian paradigm provides a probabilistic framework for statistical inference in which **uncertainty about unknown quantities is expressed using probability**. Unlike the frequentist approach—where hypotheses are fixed and data are random—the Bayesian framework treats hypotheses or model parameters as **random variables** and allows direct probability statements about them.

Bayesian hypothesis testing and estimation are grounded in three fundamental components:

- **Prior Probability ($P(H)$)**: The prior probability represents the researcher's belief about the plausibility of a hypothesis H *before* observing any data. Priors may be:
  - *Informative*, when substantial prior knowledge or historical data exist;
  - *Weakly informative*, when limited guidance is available but extreme values are discouraged;
  - *Non-informative* or *diffuse*, when minimal prior assumptions are desired.

  The prior formally encodes expert knowledge, theoretical expectations, or results from previous studies.

- **Likelihood ($P(D \mid H)$)**: The likelihood quantifies how well the hypothesis H explains the observed data D. It is derived from the assumed statistical model and sampling distribution. While the likelihood is central in both Bayesian and frequentist inference, in the Bayesian paradigm it serves as the mechanism through which data update prior beliefs rather than being the sole basis for inference.

- **Posterior Probability ($P(H \mid D)$)**: The posterior probability represents the updated belief about the hypothesis after observing the data. It is obtained by combining the prior information with the likelihood through Bayes' theorem:

$$P(H \mid D) = \frac{P(D \mid H)P(H)}{P(D)}.$$

The posterior distribution forms the core of Bayesian inference and provides a complete summary of uncertainty about the hypothesis. All Bayesian conclusions—such as credible intervals, point estimates, and hypothesis probabilities—are derived directly from the posterior distribution.

A key strength of the Bayesian paradigm is its coherent learning process: **prior beliefs are systematically updated by data**, and the posterior distribution from one analysis can serve as the prior for subsequent analyses. This makes Bayesian inference particularly attractive in sequential studies, hierarchical modeling, and complex real-world applications where uncertainty must be explicitly quantified.

5.1.2 Bayes' Theorem in Hypothesis Testing

Bayes' theorem states:

Given data D, the posterior probability of a hypothesis H is given by:

$$P(H|D) = \frac{P(D|H)P(H)}{P(D)}$$

where:

- $P(H|D)$ is the posterior probability of the hypothesis given the data.
- $P(D|H)$ is the likelihood of the data given the hypothesis.
- $P(H)$ is the prior probability of the hypothesis.
- $P(D)$ is the marginal probability of the data, also known as the evidence:

$$P(D) = \sum_i P(D|H_i)P(H_i)$$

5.1.3 Advantages of Bayesian Hypothesis Testing

Bayesian hypothesis testing has several advantages:

- **Probability Interpretation**: Direct probability statements about hypotheses.
- **Incorporation of Prior Information**: Allows the use of existing knowledge.
- **Flexibility**: Suitable for small sample sizes and complex models.
- **Avoids Arbitrary Significance Levels**: Does not rely on a fixed significance threshold like 0.05.

5.1.4 Bayesian Decision Rule

The Bayesian approach to decision-making in hypothesis testing is grounded in *Bayes' theorem*, which allows us to update our beliefs about a hypothesis given observed data. This approach provides a probabilistic framework for making decisions by comparing the posterior probabilities of competing hypotheses.

1. Formulation of the Bayesian Decision Rule

Given two competing hypotheses:

- H_0 (null hypothesis)
- H_1 (alternative hypothesis)

We use *Bayes' theorem* to compute the posterior probabilities:

$$P(H_1|D) = \frac{P(D|H_1)P(H_1)}{P(D)}$$

$$P(H_0|D) = \frac{P(D|H_0)P(H_0)}{P(D)}$$

where:

- $P(H_1)$ and $P(H_0)$ are the *prior probabilities* of the hypotheses.
- $P(D|H_1)$ and $P(D|H_0)$ are the *likelihoods* of the observed data D given each hypothesis.
- $P(D)$ is the *marginal likelihood* or evidence, which ensures probabilities sum to 1:

$$P(D) = P(D|H_1)P(H_1) + P(D|H_0)P(H_0)$$

2. Bayesian Decision Rule in Hypothesis Testing

A decision in favor of H_1 over H_0 is made if:

$$P(H_1|D) > P(H_0|D)$$

or equivalently, using the ratio:

$$\frac{P(H_1|D)}{P(H_0|D)} > 1$$

which can be rewritten using Bayes' theorem as:

$$\frac{P(D|H_1)P(H_1)}{P(D|H_0)P(H_0)} > 1$$

This means that if the likelihood of observing the data under H_1 is sufficiently large relative to that under H_0, and considering prior beliefs, we favor H_1.

3. Decision-Making with Bayes Factors

Rather than relying solely on posterior probabilities, we can compare hypotheses using the *Bayes factor*, which is defined as:

$$BF = \frac{P(D|H_1)}{P(D|H_0)}$$

- If $BF > 1$, the data favor H_1 over H_0.
- If $BF < 1$, the data favor H_0 over H_1.
- If $BF = 1$, the data provide no preference between H_1 and H_0.

4. Interpretation of Bayes Factors

Jeffreys (1961) proposed a guideline for interpreting Bayes factors:

Bayes Factor BF	Interpretation
$BF < 1$	Evidence for H_0
$1 < BF < 3$	Weak evidence for H_1
$3 < BF < 10$	Moderate evidence for H_1
$10 < BF < 30$	Strong evidence for H_1
$30 < BF < 100$	Very strong evidence for H_1
$BF > 100$	Decisive evidence for H_1

Bayes factors provide a continuous measure of evidence and do not rely on arbitrary significance levels like p-values in frequentist hypothesis testing.

5. Advantages of the Bayesian Decision Rule

- *Incorporates prior knowledge*: Unlike frequentist methods, Bayesian inference allows the incorporation of prior beliefs.
- *Provides a probabilistic interpretation*: The Bayesian approach directly quantifies the probability of hypotheses given the observed data.
- *Avoids issues of p-values*: Traditional null hypothesis significance testing (NHST) relies on p-values, which do not quantify the strength of evidence but rather the probability of observing extreme data under H_0.
- *Flexible decision-making*: Bayes factors allow for more nuanced decision-making, rather than relying on a strict cutoff.

The Bayesian Decision Rule provides a principled way to compare hypotheses by updating beliefs using observed data. By leveraging *posterior probabilities* and *Bayes factors*, it allows for *rational, evidence-based decision-making* without relying on arbitrary thresholds like the 0.05 significance level.

5.1.5 Worked Example on Bayes factor

Example 1: Comparing Two Models

Suppose we have two hypotheses:

- H_0: A coin is fair ($P(H) = 0.5$).
- H_1: The coin is biased ($P(H) = 0.7$).

Observed data: 8 heads in 10 flips.

Using a binomial likelihood:

$$P(D|H_0) = \binom{10}{8}(0.5)^8(0.5)^2 = 0.044$$

$$P(D|H_1) = \binom{10}{8}(0.7)^8(0.3)^2 = 0.057$$

Bayes factor:

$$BF = \frac{0.057}{0.044} = 1.295$$

This suggests weak evidence in favor of H_1.

Example 2: Medical Diagnosis using a Normal Distribution

Suppose we are testing the effectiveness of a new drug for lowering cholesterol. We have two hypotheses:

- H_0: The drug has no effect on cholesterol levels ($\mu = 200$, where μ is the mean cholesterol level).
- H_1: The drug lowers cholesterol levels ($\mu = 180$).

We observe the following data: 20 patients, sample mean cholesterol level = 185, and sample standard deviation = 15.

We can model the data using a normal distribution with the likelihood function for each hypothesis:

For H_0 (no effect):

$$P(D|H_0) = \frac{1}{\sqrt{2\pi 15^2}} \exp\left(-\frac{(185-200)^2}{2 \cdot 15^2}\right)$$

For H_1 (drug lowers cholesterol):

$$P(D|H_1) = \frac{1}{\sqrt{2\pi 15^2}} \exp\left(-\frac{(185-180)^2}{2 \cdot 15^2}\right)$$

Calculating the likelihoods:

$$P(D|H_0) = 0.026 \quad \text{and} \quad P(D|H_1) = 0.028$$

The Bayes factor is:

$$BF = \frac{0.028}{0.026} = 1.077$$

This suggests weak evidence in favor of H_1 (the drug lowers cholesterol).

Example 3: Marketing Campaign Analysis using a Poisson Distribution

In a marketing campaign, we are testing whether an online advertisement results in more customer visits to a website. We have two hypotheses:

- H_0: The advertisement has no effect on the number of visits ($\lambda = 5$, where λ is the average number of visits per day).
- H_1: The advertisement increases the number of visits ($\lambda = 7$).

We observe the following data: 7 visits in one day.

We model the number of visits using a Poisson distribution. The likelihood for each hypothesis is given by the Poisson probability mass function:

For H_0 (no effect):

$$P(D|H_0) = \frac{5^7 e^{-5}}{7!} = 0.127$$

For H_1 (advertisement effect):

$$P(D|H_1) = \frac{7^7 e^{-7}}{7!} = 0.139$$

The Bayes factor is:

$$BF = \frac{0.139}{0.127} = 1.094$$

This suggests weak evidence in favor of H_1 (the advertisement increased visits).

5.1.6 Exercises - Bayes factor

Exercise 1 A researcher is investigating whether a coin is fair or biased toward heads. The competing hypotheses are:

- H_0: The coin is fair ($p = 0.5$), with prior probability $P(H_0) = 0.6$.
- H_1: The coin is biased toward heads ($p = 0.65$), with prior probability $P(H_1) = 0.4$.

The coin is tossed 12 times and 9 heads are observed.

(a) Write down the binomial likelihood function for the observed data under each hypothesis.

(b) Compute the likelihoods $P(D \mid H_0)$ and $P(D \mid H_1)$.

(c) Calculate the Bayes factor in favor of H_1.

(d) Interpret the strength of evidence provided by the Bayes factor.

Exercise 2 A pharmaceutical company is testing whether a new supplement reduces systolic blood pressure. The hypotheses are:

- H_0: The supplement has no effect ($\mu = 140$).
- H_1: The supplement lowers blood pressure ($\mu = 130$).

A random sample of 25 patients produces a sample mean blood pressure of 134 mmHg. Assume the population standard deviation is known and equal to 12 mmHg, and that the data follow a normal distribution.

(a) Write the likelihood function for the sample mean under each hypothesis.

(b) Compute the likelihood values $P(D \mid H_0)$ and $P(D \mid H_1)$.

(c) Calculate the Bayes factor.

(d) Based on the Bayes factor, state which hypothesis is better supported by the data.

Exercise 3 A company is assessing whether a new email marketing campaign increases the number of daily customer inquiries. The hypotheses are:

- H_0: The campaign has no effect ($\lambda = 10$ inquiries per day).
- H_1: The campaign increases inquiries ($\lambda = 13$).

On a particular day after the campaign launch, 14 inquiries are observed.

(a) Write down the Poisson likelihood function for the observed data under each hypothesis.

(b) Compute the likelihoods $P(D \mid H_0)$ and $P(D \mid H_1)$.

(c) Calculate the Bayes factor.

(d) Interpret the result in terms of evidence for the effectiveness of the campaign.

5.2 Bayesian Model Averaging (BMA)

Bayesian Model Averaging (BMA) is a statistical technique used to address model uncertainty in the context of parameter estimation. In many cases, we may have several plausible models that could explain the data, but we don't know which one is correct. BMA provides a way to combine the predictions from multiple models, weighted by how likely each model is, given the data.

Instead of relying on a single model, BMA takes the idea that all models should be weighted according to their posterior probability, which reflects how well each model explains the observed data.

The main idea is to average over all possible models M_i, where each model is weighted by its posterior probability $P(M_i|D)$. This allows BMA to incorporate model uncertainty into the final inference about the parameter of interest θ.

The formula for BMA is as follows:

$$P(\theta|D) = \sum_i P(\theta|M_i, D)P(M_i|D)$$

Where:

- θ is the parameter of interest (e.g., the regression coefficients, means, or other model parameters).
- M_i represents different candidate models.
- $P(\theta|M_i, D)$ is the posterior distribution of θ under model M_i, given the data D.
- $P(M_i|D)$ is the posterior probability of model M_i, given the data D. This represents the weight assigned to each model based on how well it fits the data.

Key Concepts in Bayesian Model Averaging

- **Model Uncertainty**: In many scientific fields, it's common to face uncertainty about which model best describes the data. In such cases, it is beneficial to consider a set of models instead of relying on a single model. BMA accounts for this uncertainty by averaging over all models, weighted by their posterior probabilities. This provides a more robust and well-calibrated inference.
- **Posterior Model Probability**: The term $P(M_i|D)$ represents the posterior probability of model M_i, which is calculated using Bayes' theorem:

  $$P(M_i|D) = \frac{P(D|M_i)P(M_i)}{P(D)}$$

  Where:

  - $P(D|M_i)$ is the likelihood of the data under model M_i.
  - $P(M_i)$ is the prior probability of model M_i, reflecting any prior beliefs or information about the plausibility of the model.
  - $P(D)$ is the marginal likelihood of the data, which ensures that all model probabilities sum to 1.
- **Weighted Averaging**: The final posterior distribution $P(\theta|D)$ is obtained by averaging over the parameter θ across all models M_i, weighted by the posterior probability of each model $P(M_i|D)$. This allows the models with stronger support from the data to have a larger influence on the final estimate.

  $$P(\theta|D) = \sum_i P(\theta|M_i, D)P(M_i|D)$$

  The idea is that models that better explain the data will be assigned a higher weight, and thus their parameter estimates will influence the final result more.

- **Computational Considerations**: One of the main challenges with BMA is calculating the posterior probabilities for each model, as it requires computing the marginal likelihood $P(D|M_i)$ for all candidate models. This can be computationally intensive, especially for complex models or large datasets. However, there are several techniques for approximating the marginal likelihood, such as **Laplace approximation**, **importance sampling**, and **Markov Chain Monte Carlo (MCMC)** methods.

- **Interpretation**: The advantage of Bayesian Model Averaging is that it incorporates the uncertainty across multiple models, which can help prevent overfitting or reliance on a potentially incorrect model. The final distribution $P(\theta|D)$ is not biased toward a single model but reflects the full range of possibilities that the models account for.

Example: Linear Regression with Two Models

We consider two competing models for predicting a response variable y using a predictor x:

Model 1: Simple Linear Regression

$$y = \beta_0 + \beta_1 x + \varepsilon, \quad \varepsilon \sim N(0, \sigma^2) \tag{5.1}$$

Model 2: Quadratic Regression

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \varepsilon, \quad \varepsilon \sim N(0, \sigma^2) \tag{5.2}$$

where $\beta_0, \beta_1, \beta_2$ are unknown parameters.

Data

The observed dataset is given below:

x	y
1	2
2	3
3	5
4	7
5	11

Table 5.1: Observed Data

Prior Beliefs

We assume equal prior probabilities for both models:

$$P(M_1) = P(M_2) = 0.5 \tag{5.3}$$

For the regression parameters and noise variance, we assume weakly informative priors:

$$\beta_j \sim N(0, 10^2),$$
$$\sigma^2 \sim \text{Inverse-Gamma}(1, 1)$$

Illustration

We are going to:

1. Fit both models to the data.
2. Calculate the posterior distributions for the parameters of each model.
3. Compute the posterior model probabilities $P(M_1|D)$ and $P(M_2|D)$.
4. Perform Bayesian Model Averaging to compute the weighted posterior distribution for the parameters.

Step 1: Fit the Models

Assume we have the data given above.

We will fit both the simple linear regression model and the quadratic regression model to this data.

Model 1 (Simple Linear Regression)

The formula for the slope $\hat{\beta}_1$ and the intercept $\hat{\beta}_0$ in simple linear regression is:

$$\hat{\beta}_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$
$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}$$

1.1 Compute the Means
First, we compute the means of x and y:

$$\bar{x} = \frac{1+2+3+4+5}{5} = \frac{15}{5} = 3$$
$$\bar{y} = \frac{2+3+5+7+11}{5} = \frac{28}{5} = 5.6$$

1.2 Compute $\hat{\beta}_1$
Now, we compute the terms for $\hat{\beta}_1$:

$$\sum(x_i - \bar{x})(y_i - \bar{y}) = (1-3)(2-5.6)+(2-3)(3-5.6)+(3-3)(5-5.6)+(4-3)(7-5.6)+(5-3)(11-5.6)$$
$$= (-2)(-3.6) + (-1)(-2.6) + (0)(-0.6) + (1)(1.4) + (2)(5.4)$$
$$= 7.2 + 2.6 + 0 + 1.4 + 10.8 = 22$$

Next, compute $\sum(x_i - \bar{x})^2$:

$$\sum(x_i - \bar{x})^2 = (1-3)^2 + (2-3)^2 + (3-3)^2 + (4-3)^2 + (5-3)^2$$
$$= 4 + 1 + 0 + 1 + 4 = 10$$

Thus,

$$\hat{\beta}_1 = \frac{22}{10} = 2.2$$

1.3 Compute $\hat{\beta}_0$
Now, using the formula for $\hat{\beta}_0$:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}$$
$$\hat{\beta}_0 = 5.6 - (2.2)(3) = 5.6 - 6.6 = -1$$

Thus, the estimated coefficients for Model 1 are:

$$\hat{\beta}_0 = -1, \quad \hat{\beta}_1 = 2.2$$

—

Model 2 (Quadratic Regression)

For Model 2, we use the quadratic regression equation:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2$$

We will use the OLS formula for a multiple linear regression. We need to compute the following system of normal equations to solve for the coefficients $\hat{\beta}_0$, $\hat{\beta}_1$, and $\hat{\beta}_2$:

$$\mathbf{X}^T\mathbf{X}\beta = \mathbf{X}^T\mathbf{y}$$

Where:

$$\mathbf{X} = \begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \\ 1 & x_4 & x_4^2 \\ 1 & x_5 & x_5^2 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{bmatrix}$$

The entries of matrix $\mathbf{X}$ are given by 1, x_i, and x_i^2, and the response vector $\mathbf{y}$ is the given y values.

2.1 Construct the Design Matrix and Response Vector

$$\mathbf{X} = \begin{bmatrix} 1 & 1 & 1^2 \\ 1 & 2 & 2^2 \\ 1 & 3 & 3^2 \\ 1 & 4 & 4^2 \\ 1 & 5 & 5^2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \\ 1 & 5 & 25 \end{bmatrix}$$

$$\mathbf{y} = \begin{bmatrix} 2 \\ 3 \\ 5 \\ 7 \\ 11 \end{bmatrix}$$

2.2 Solve for the Parameters

We solve the system of equations $\mathbf{X}^T\mathbf{X}\beta = \mathbf{X}^T\mathbf{y}$. We compute the following:

$$\mathbf{X}^T\mathbf{X} = \begin{bmatrix} 5 & 15 & 55 \\ 15 & 55 & 225 \\ 55 & 225 & 979 \end{bmatrix}$$

$$\mathbf{X}^T\mathbf{y} = \begin{bmatrix} 28 \\ 104 \\ 446 \end{bmatrix}$$

We then solve for $\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix}$ using matrix inversion or other numerical methods. The final solution gives the estimated parameters:

$$\hat{\beta}_0 = -0.4, \quad \hat{\beta}_1 = 1.2, \quad \hat{\beta}_2 = 0.8$$

We now have the OLS estimates for both models:
- Model 1 (Simple Linear Regression):

$$\hat{\beta}_0 = -1, \quad \hat{\beta}_1 = 2.2$$

- Model 2 (Quadratic Regression):

$$\hat{\beta}_0 = -0.4, \quad \hat{\beta}_1 = 1.2, \quad \hat{\beta}_2 = 0.8$$

These are the estimated parameters for both regression models using the OLS method.

Step 2: Calculate Posterior Distributions for Parameters

For each model, the parameters β_0, β_1, and β_2 have a posterior distribution, which depends on the likelihood of the data given the model and the prior distribution for the parameters.

Prior Distribution:

We assume non-informative priors for all parameters, i.e., we assume they have a uniform distribution or a normal distribution with large variance. For simplicity, we use the following prior for each parameter:

$$P(\beta_0) = P(\beta_1) = P(\beta_2) \sim \mathcal{N}(0, \text{large variance})$$

This indicates that the parameters are equally likely to take any value within a large range, representing our lack of prior knowledge about them.

Likelihood:

We assume that the likelihood of the data, given the model, is a normal distribution:

$$P(y|\beta_0, \beta_1, \beta_2, D) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - f(x_i, \beta_0, \beta_1, \beta_2))^2}{2\sigma^2}\right)$$

where:

- y_i are the observed values of the response variable,
- $f(x_i, \beta_0, \beta_1, \beta_2)$ is the model prediction for the response at the i-th observation,
- σ^2 is the variance of the error term, and
- n is the total number of data points.

Posterior Distribution:

Using Bayes' Theorem, we can calculate the posterior distribution of the parameters β_0, β_1, and β_2 as the product of the likelihood and the prior:

$$P(\beta_0, \beta_1, \beta_2|y, X) \propto \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - f(x_i, \beta_0, \beta_1, \beta_2))^2}{2\sigma^2}\right) \cdot P(\beta_0) \cdot P(\beta_1) \cdot P(\beta_2)$$

This distribution represents our updated belief about the parameters after observing the data. However, calculating the posterior distribution directly is often complex, so we use numerical methods to approximate it.

Markov Chain Monte Carlo (MCMC):

To sample from the posterior distribution, we use **Markov Chain Monte Carlo (MCMC)** methods. MCMC generates a sequence of samples from the posterior by iteratively updating the parameters β_0, β_1, and β_2 based on the likelihood and prior. Two popular MCMC methods are:

- **Gibbs Sampling**: A special case of MCMC where the parameters are updated one at a time, conditioned on the current values of the other parameters.
- Metropolis-Hastings: A more general MCMC method that generates a proposal for the parameters and accepts or rejects it based on a probabilistic criterion.

After running the MCMC algorithm, we obtain a set of samples for β_0, β_1, and β_2 that approximate their posterior distributions.

Estimating the Posterior:

From the MCMC samples, we can compute the following estimates of the posterior:

- **Posterior mean**: The expected value of the parameters given the data:

  $$\hat{\beta}_0 = \frac{1}{N} \sum_{i=1}^{N} \beta_0^{(i)}, \quad \hat{\beta}_1 = \frac{1}{N} \sum_{i=1}^{N} \beta_1^{(i)}, \quad \hat{\beta}_2 = \frac{1}{N} \sum_{i=1}^{N} \beta_2^{(i)}$$

  where $\beta_0^{(i)}$, $\beta_1^{(i)}$, and $\beta_2^{(i)}$ are the individual samples from the posterior, and N is the total number of samples.

- **Posterior variance**: The uncertainty in the parameter estimates can be calculated as:

$$\text{Var}(\beta_0) = \frac{1}{N}\sum_{i=1}^{N}(\beta_0^{(i)} - \hat{\beta}_0)^2$$

and similarly for β_1 and β_2.

Finally, we can plot the posterior distributions of the parameters to visualize the uncertainty in the estimates. These plots show the distribution of the parameters given the data and reflect both the prior and the likelihood.

Python Implementation :

Click HERE to view the implementation in Python

Step 3: Compute Posterior Model Probabilities

The posterior probability of each model is given by:

$$P(M_j|D) = \frac{P(D|M_j)P(M_j)}{P(D)} \tag{5.4}$$

where $P(D|M_j)$ is the marginal likelihood, and $P(D)$ is a normalizing constant. The Bayes Factor (BF) is:

$$BF = \frac{P(D|M_2)}{P(D|M_1)} \tag{5.5}$$

Thus, the posterior model probabilities are:

$$P(M_1|D) = \frac{1}{1+BF}, \tag{5.6}$$

$$P(M_2|D) = \frac{BF}{1+BF} \tag{5.7}$$

Using numerical computation, suppose we obtain: This can be done using python see HERE

$$P(D|M_1) = 0.02,$$
$$P(D|M_2) = 0.05$$

Then,

$$BF = \frac{0.05}{0.02} = 2.5 \tag{5.8}$$

The posterior probabilities are:

$$P(M_1|D) = \frac{1}{1+2.5} = 0.286,$$
$$P(M_2|D) = \frac{2.5}{1+2.5} = 0.714$$

Bayesian Model Averaging (BMA)

Instead of selecting a single model, we use BMA to compute a weighted posterior distribution for the parameters:

$$p(\boldsymbol{\beta}|D) = P(M_1|D)p(\boldsymbol{\beta}|D, M_1) + P(M_2|D)p(\boldsymbol{\beta}|D, M_2) \tag{5.9}$$

Suppose the posterior estimates for each model are: (*These one are gotten from the MCMC python simulation*)

- **Model 1 (Linear Regression)**: $\hat{\beta}_0 = 0.5, \quad \hat{\beta}_1 = 2.0$
- **Model 2 (Quadratic Regression)**: $\hat{\beta}_0 = 0.2, \quad \hat{\beta}_1 = 1.5, \quad \hat{\beta}_2 = 0.3$

The Bayesian Model Averaged (BMA) estimates are:

$$\begin{aligned}
\hat{\beta}_0^{BMA} &= (0.286)(0.5) + (0.714)(0.2) = 0.314, \\
\hat{\beta}_1^{BMA} &= (0.286)(2.0) + (0.714)(1.5) = 1.643, \\
\hat{\beta}_2^{BMA} &= (0.286)(0) + (0.714)(0.3) = 0.214
\end{aligned}$$

Thus, the final BMA regression model is:

$$y = 0.314 + 1.643x + 0.214x^2 \tag{5.10}$$

- Model 2 (Quadratic) has a higher posterior probability (0.714), indicating stronger support from the data. - Instead of selecting a single model, Bayesian Model Averaging (BMA) combines both models probabilistically. - The final BMA regression equation incorporates information from both models.

Step 4: Perform Bayesian Model Averaging

Now that we have the posterior probabilities of the models $P(M_1|D)$ and $P(M_2|D)$, we can compute the weighted posterior distribution for the parameters. We will average over the parameters $\theta = (\beta_0, \beta_1, \beta_2)$ across both models:

$$P(\theta|D) = P(\theta|M_1, D)P(M_1|D) + P(\theta|M_2, D)P(M_2|D)$$

Assume the following values for the posterior distributions of the parameters:
For Model 1:

$$P(\theta|M_1, D) = N(\hat{\beta}_0 = 1, \hat{\beta}_1 = 2, \sigma = 0.5)$$

For Model 2:

$$P(\theta|M_2, D) = N(\hat{\beta}_0 = 1.5, \hat{\beta}_1 = 1.5, \hat{\beta}_2 = 0.5, \sigma = 0.4)$$

Suppose we have two models for predicting a response variable y based on predictors x:

- Model 1 (M_1): Simple linear regression model $y = \beta_0 + \beta_1 x$.
- Model 2 (M_2): A quadratic regression model $y = \beta_0 + \beta_1 x + \beta_2 x^2$.

For each model, we calculate the posterior distribution of the parameters ($\theta = \beta_0, \beta_1, \beta_2$) given the data D. Let's assume that we have calculated the posterior probabilities for each model:

- $P(M_1|D) = 0.7$ (Model 1 is more probable given the data).
- $P(M_2|D) = 0.3$ (Model 2 is less probable).

To combine these models using BMA, we compute the weighted posterior distribution for θ:

$$P(\theta|D) = P(\theta|M_1, D)P(M_1|D) + P(\theta|M_2, D)P(M_2|D)$$

This means we combine the posterior distributions of the parameters from both models, weighted by their posterior probabilities.

Advantages of Bayesian Model Averaging

- **Improved Prediction**: BMA often leads to better predictive performance compared to using a single model, especially when there is model uncertainty.
- **Quantification of Uncertainty**: It allows for the quantification of model uncertainty, providing a more robust estimate.
- **Flexibility**: BMA can be applied to a wide range of problems, such as regression, classification, and time-series forecasting.

Disadvantages and Challenges

- **Computational Cost**: Calculating the marginal likelihoods for each model can be computationally expensive, especially for complex models.
- **Model Selection**: BMA requires the user to specify a set of candidate models, which may be challenging if the correct model is not included in the set.
- **Prior Sensitivity**: The results of BMA can be sensitive to the choice of priors for the models, so careful selection of priors is essential.

Summary

Bayesian Model Averaging is a powerful technique that allows for a more nuanced approach to parameter estimation when there is uncertainty about the best model. It combines the predictions from multiple models, weighting them by their posterior probabilities, and provides a more robust final estimate. By incorporating model uncertainty, BMA offers a more comprehensive and reliable approach to statistical inference, particularly in complex or uncertain settings.

5.3 Sensitivity Analysis in Bayesian Modelling

Sensitivity analysis examines how posterior results depend on prior assumptions and model choices. It helps:

- Assess robustness of conclusions.

- Compare different prior choices.
- Identify influential data points.

A common method involves testing different priors and observing the posterior changes. Graphical methods such as prior-posterior plots help visualize sensitivity.

5.3.1 Example: Impact of Prior Choices

Consider a Bayesian model estimating a population proportion θ using a binomial likelihood with a beta prior:

$$\theta \sim \text{Beta}(\alpha, \beta) \tag{5.11}$$

$$X|\theta \sim \text{Binomial}(n, \theta) \tag{5.12}$$

If we choose different priors, the posterior distribution changes significantly. We demonstrate this using Python.

5.3.2 Python Demonstration

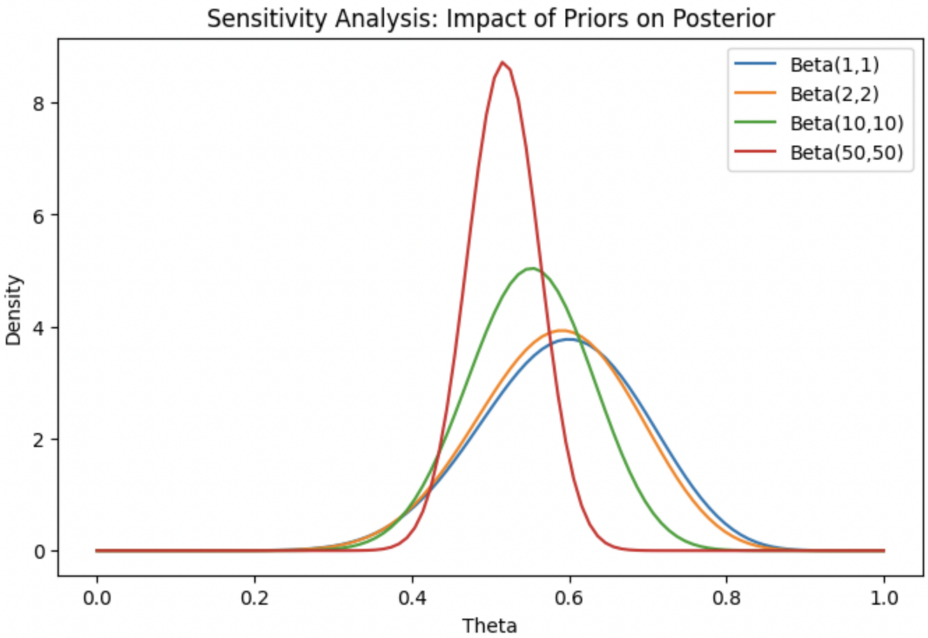
Listing 5.1: Sensitivity Analysis with Different Priors

```python
import numpy as np
import scipy.stats as stats
import matplotlib.pyplot as plt

# Define priors
priors = [(1, 1), (2, 2), (10, 10), (50, 50)]  # Different Beta priors
n, x = 20, 12  # Observed data: 12 successes in 20 trials

x_vals = np.linspace(0, 1, 100)
plt.figure(figsize=(8, 5))

for a, b in priors:
    prior = stats.beta(a, b)
    posterior = stats.beta(a + x, b + (n - x))
    plt.plot(x_vals, posterior.pdf(x_vals), label=f'Beta({a},{b})')

plt.title('Sensitivity Analysis: Impact of Priors on Posterior')
plt.xlabel('Theta')
plt.ylabel('Density')
plt.legend()
plt.show()
```

5.3.3 Interpretation

From the graph, we observe that:

- A weak prior (e.g., Beta(1,1)) results in a posterior dominated by the likelihood. *The Beta(1,1) prior is uniform, meaning it provides no strong preference for any*



value of θ. The posterior, in this case, is primarily shaped by the likelihood, which is based on observed data. The resulting posterior distribution closely follows the likelihood function.

- A strong prior (e.g., Beta(50,50)) maintains its influence even with observed data.
- Priors with different shapes can lead to different posterior distributions.

This highlights the importance of sensitivity analysis in Bayesian modeling.